

NebuLink Gateway NLG7644

Datasheet



System Level Solutions, Inc. (USA)
511 N. Washington Avenue
Marshall, Texas - 75670
(408) 852 - 0067

<https://www.slscorp.com>

Document Version: 2.0
Document Date: June 2021

Copyright ©2021, System Level Solutions, Inc. (SLS) All rights reserved. SLS, an Embedded systems company, the stylized SLS logo, Nebulae logo specific device designations, and all other words and logos that are identified as trademarks and/or service marks are, unless noted otherwise, the trademarks and service marks of SLS in India and other countries. All other products or service names are the property of their respective holders. SLS products are protected under numerous U.S. and foreign patents and pending applications, mask working rights, and copyrights. SLS reserves the right to make changes to any products and services at any time without notice. SLS assumes no responsibility or liability arising out of the application or use of any information, products, or service described herein except as expressly agreed to in writing by SLS. SLS customers are advised to obtain the latest version of specifications before relying on any published information and before orders for products or services.

ds_plora4ggtw_2.0

Revision History

Table below shows the revision history of NLG7644 Gateway datasheet.

Version	Date	Description
2.0	June 2021	- Updated details as per new naming convention NLG7644 and features - Added Chp 3 Mounting Guidelines
1.1	October 2020	- Added Information to Nebulae in About this Manual section - Updated Transmit Power to 20 dBm in Table 1-1 - Added Chp 2 Wiring Diagram
1.0	September 2020	First publication

How to Contact SLS

For the most up-to-date information about SLS products, go to the SLS worldwide website at <https://www.slscorp.com>. For additional information about SLS products, consult the source shown below.

Information Type	E-mail
Product literature services, SLS literature services, Non-technical customer services, Technical support.	support@slscorp.com

Information to Nebulae

Nebulae[®] is a business vertical of SLS to provide Internet of Things (IoT) solutions. For the most up-to-date information about Nebulae products and solutions, go to Nebulae worldwide website at <https://www.nebulae.io>. For additional information, contact at sales@nebulae.io.

Typographic Conventions

This datasheet uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics. Examples: <i>Figure 1-1. NLG7644 Gateway, Table 1-1. NLG7644 Gateway Specifications.</i>
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.
	The hand points to special information that requires special attention
	The caution indicates required information that needs special consideration and understanding and should be read prior to starting or continuing with the procedure or process.
	The warning indicates information that should be read prior to starting or continuing the procedure or processes.
	The feet direct you to more information on a particular topic.

<i>About this Manual</i>	<i>iii</i>
Revision History	iii
How to Contact SLS	iii
Information to Nebulae	iii
Typographic Conventions	iv
1. Overview	1
General Description	1
Features	2
Main Board	2
Enclosure	2
Accessories	3
Benefits	3
Hardware Interface	4
Applications	4
Block Diagram	5
Specifications	6
2. Wiring Diagram	9
3. Mounting Guidelines	10
Pole Mounting	10
Wall Mounting	11
4. Getting Started	14
LED Indication	14
Getting Started	14

General Description

Gateways are key components for IoT commercial deployment which helps to remotely manage and control the solutions. SLS offers various gateway solutions to meet market needs. LoRa® is a well known for long range communication and SLS has now increase their portfolio by adding the NLG7644 - A LoRaWAN gateway. This gateway is pre-programmed with full featured LoRaWAN™ network base station. This gateway helps the LoRa devices to connect over 4G to cloud. It has a range of over 15 KM line-of-sight and over 2 KM in dense urban environments.

NLG7644 gateway brings more flexibility for the developer to create an enterprise grade solution with the help of our flexible development support structure which allows faster development and time to market.

Figure 1-1. NLG7644 Gateway



Features

Main Board

- Processor
 - ARM Cortex™ A7
 - Frequency 1.2 GHz
- Storage Memory:
 - 256MB RAM (40MB for End Application)
 - 512MB Flash (20MB for End Application)
 - Upto 32GB microSD
- LoRaWAN Concentrator
 - Standard version with 8 channel Gateway
 - Tx Power: configurable from +5 to +27 dBm
 - Rx sensitivity: down to -137 dBm
- Communication Interface
 - Cellular:
 - FDD-LTE: B1/B2/B3/B5/B7/B8/B20
 - TDD-LTE: B38/B40
 - HSPA/UMTS:900/850/1800/1900/2100MHz
 - GSM/GPRS/EDGE850/900/1800/1900MHz
 - Ethernet 1 x IEEE 802.3 10/100M/1Gbps
 - GPS: Available

Enclosure

- NEMA rating: 1, 2, 3, 3R, 3RX, 3S, 3SX, 3X, 4, 4X, 5, 6, 6P, 12, 13
- IP65, IP66, IP67 and IP68 rating
- Flame rating: UL94 5VA
- Interface: 3 x N-Type connectors for Antenna, 1 x Ethernet port
- Weight: 800 gm
- Dimensions: 208.17 x 190 x 60.20 mm

Accessories

- Pole Mounting Kit
 - Pole mounting clamp GI 1.6mm
 - Hose clamp for 70 -150mm diameter
 - M5 nut (2 nos.)
 - M5 X20mm screw (2 nos.)
 - Metal Washer (2 nos.)
- Wall mounting Kit
 - 32 x 6mm self tapper pan screw with PVC grips (4 nos.)
- GPS Antenna
- LoRaWAN Antenna
- LTE Antenna

Benefits

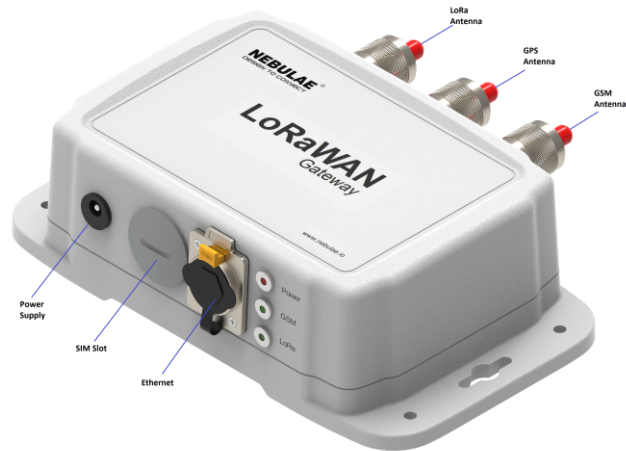
Following are the benefits of NLG7644 Gateway.

- Enterprise grade network gateway with your own configuration
- LoRaWAN™ Stack Inside and integrate the Web UI for management
- Complete Hardware specification including LoRa concentrator, Cellular, and GPS
- IP67 waterproof enclosure

Hardware Interface

Figure 1-2. shows the hardware interface of NLG7644 Gateway.

Figure 1-2. Hardware Interface of LoRaWAN 4G Gateway



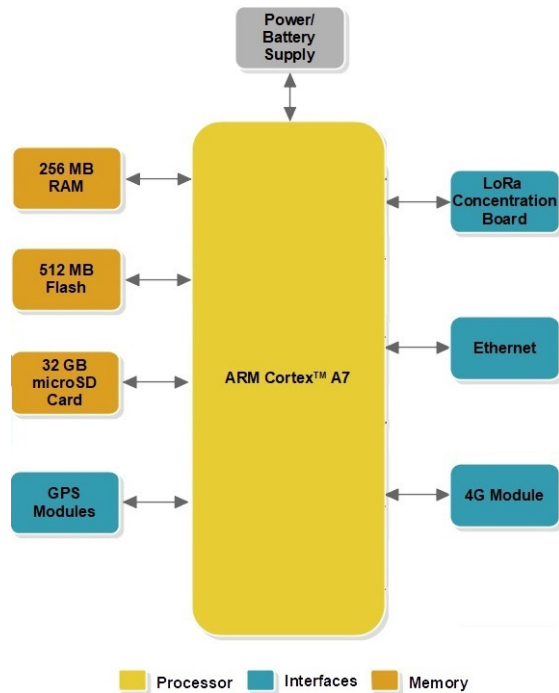
Applications

- Industry, Defence, Military Applications
- Smart Metering
- Street Lighting
- Smart Buildings & Home Automation
- Logistics and Supply Chain Management
- Smart Cities
- Smart Factory
- Data Centers/Server Rooms

Block Diagram

Figure 1-3. shows NLG7644 Gateway block diagram.

Figure 1-3. NLG7644 Gateway Block Diagram



Specifications

Table 1-1. describes the specifications of NLG7644 Gateway.

<i>Table 1-1. LoRaWAN 4G Gateway Specifications</i>	
Processor	
Processor	ARM Cortex™ A7
Frequency	1.2 GHz
Storage	
Internal RAM	256 MB
Internal Flash	512 MB
Additional Support	upto 32 GB microSD card
Communication Interface	
GPS	Available
Serial	1 x RS485
Ethernet	1 x IEEE 802.3 10/100M/1G bps
Cellular	FDD-LTE: B1/B2/B3/B5/B7/B8/B20 TDD-LTE: B38/B40 HSPA/UMTS: 900/850/1800/1900/2100 MHz GSM/GPRS/EDGE: 850/900/1800/1900 MHz
LoRa Specification	
LoRa Module	SX1301 based, 8 channels support
Operating Frequency	EU868, IN865
Transmit Power	+5 ~ +20 dBm (Configurable)
Receiver Sensitivity	-137 dBm (Min)
Software Specification	
Stack	QuecOpen™ Linux OS running NebuLink Gateway SDK with validated Modbus RTU, Modbus TCP, DLMS Stack

Table 1-1. LoRaWAN 4G Gateway Specifications

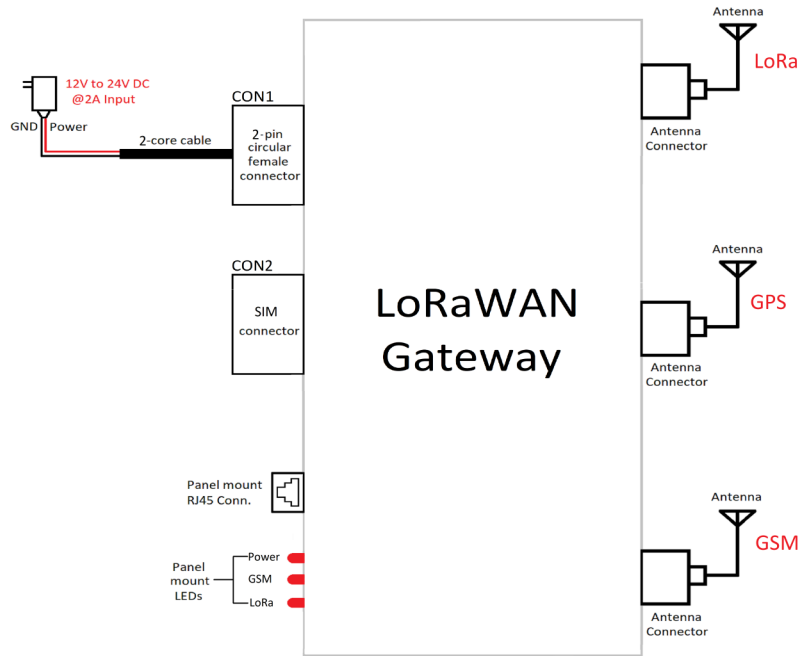
LoRa	• Supports class A, and C • Supports LoRaWAN protocol • Supports data logger • Supports location setup • Supports server address & port setup • Supports Gateway bridge with TLS encryption • Support of configurable State Packet • LoRa Frame filtering (node whitelisting) • Buffering of upto 5000 LoRa frames in case of NS outage (no data loss)
Network	• Supports uplink backup • Supports DHCP Client
Management	• Supports firmware update • Supports NTP
Electrical Specification	
Power Supply	12 - 24 VDC @4A
Supported Battery Supply	3.7/4.2 V @2A
Operating Temperature	-40 to +85 °C
Humidity	Relative humidity 15% to 93% non condensing
Safety	EMS Protection, ESD (IEC 61000-4-2) 4 kV Contact for each terminal, EFT (IEC 61000-4-4) ±2 kV for Power
Mechanical Specification	
Ingress Protection	IP67
Enclosure Material	Polycarbonate
Weight	800 gms
Dimension	208.17 x 190 x 60.20 mm
Installation Method	Pole or Wall mounting
General	
LEDs	3 x Status LED

Table 1-1. LoRaWAN 4G Gateway Specifications

Accessories	
Antennas	1 x Cellular, 1 x GPS, 1 x LoRa (Fiber glass with 5 dBi gain)
Mounting Kit	Pole Mounting Kit • Pole mounting clamp GI 1.6mm • Hose clamp for 70 -150mm diameter • M5 nut (2 nos.) • M5 X20mm screw (2 nos.) • Metal Washer (2 nos.) Wall mounting Kit • 32 x 6mm self tapper pan screw with PVC grips (4 nos.)

2. Wiring Diagram

Figure 2-1. LoRaWAN 4G Gateway External Wiring Diagram



- Refer the wire connection to connect with corresponding external peripheral devices pin.
- Position of panel mount connectors, antennas and LEDs on the enclosure may differ on gateway while connection.
- Use power supply as suggested.

3. Mounting Guidelines

This chapter guides you for mounting NLG7644 gateway on wall or pole with accessories. This shall be followed before gateway's configuration set up. Let's see the steps in details for both mounting options.

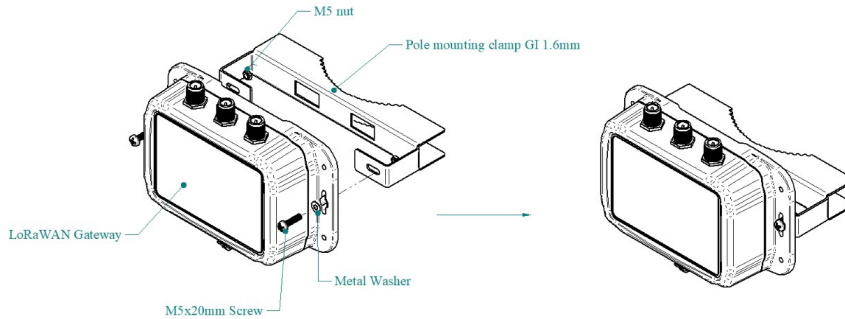
Pole Mounting

To mount gateway on pole, follow the steps as below:

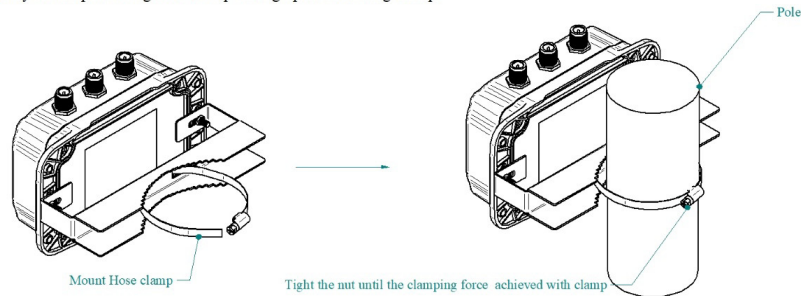
1. Connect provided pole mounting clamp with gateway using M5 X20mm screw, metal washer & M5 nut as shown in [Figure 3-1](#).
2. Once the mounting clamp is connected, use hose clamp to fix the gateway on pole. Tighten up the screw of the hose clamp to fix the gateway. See [Figure 3-1](#).

Figure 3-1. Pole Mounting Guidelines

Step 1 : Attach Pole mounting clamp GI 1.6mm with LoRaWAN Gateway Using M5x20mm screw ,nut & metal washer.

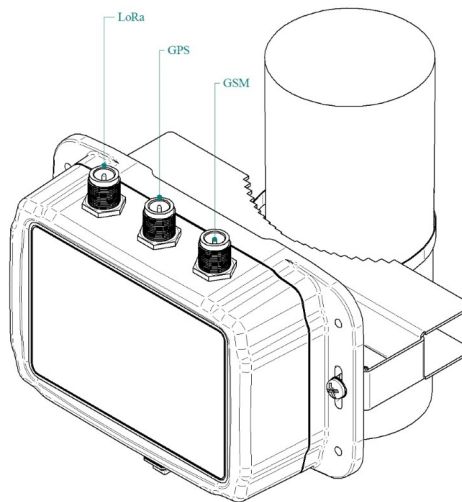


Step 2 : Mount Gateway on the pole using hose clamp through pole mounting clamp.



3. Connect the antennas at the positions indicated in [Figure 3-2](#).

Figure 3-2. Pole Mounted Gateway



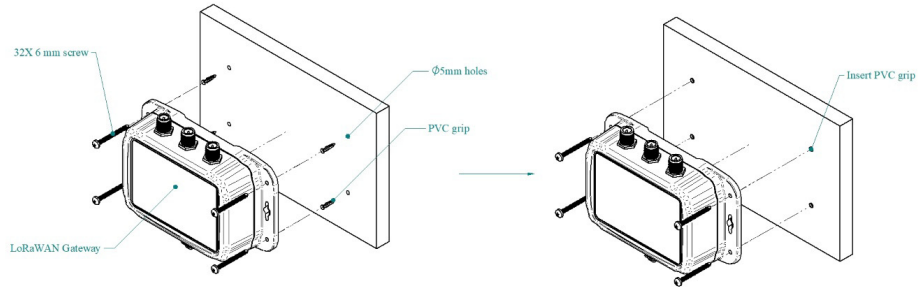
Wall Mounting

To mount the gateway on wall, follow the steps below:

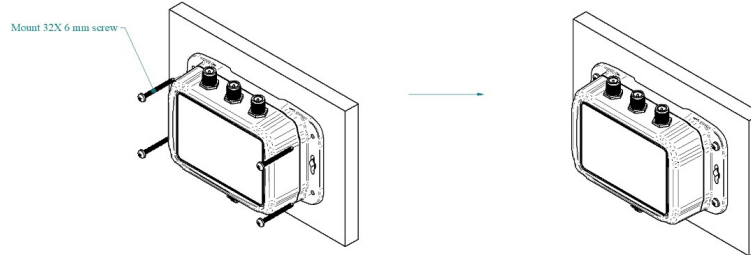
1. Drill four holes of 5mm in size on wall as per mounting hole position of gateway. See [Figure 3-3](#).
2. Insert PVC grip inside the holes.
3. Mount 32 x 6mm screw in the gateway and fix it on the wall at hole positions. See [Figure 3-3](#).

Figure 3-3. Wall Mounting Guidelines

Step 1 : Drill 5mm holes in the wall with reference to gateway mountings and fix PVC grip in it.

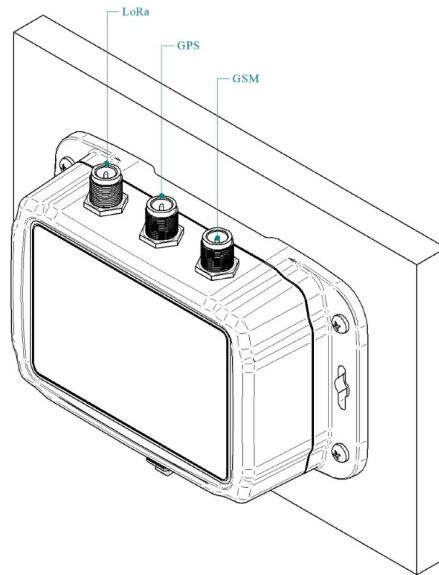


Step 2 : Use 32x6 mm self tapping CSK screw to mount gateway on the wal Locate the screws and tighten it up by screw driver set.



4. Connect the antennas at the positions indicated in [Figure 3-4](#).

Figure 3-4. Wall Mounted Gateway



NLG7644 Gateway provides three LEDs for status indication. Let's see their functions in detail.

LED Indication

There are three status LEDs. [Table 4-1](#). explains their status indication for various gateway operation.

Table 4-1. LED Status Indication

LEDs	Use For	Status	Status Indication
LED1	Power	ON	Gateway is power on
		OFF	Gateway is power off
LED2	LoRa	Flash	LoRa communication is going on
LED3	GSM/GPRS	Flash	Data transfer is going on

Getting Started

Turn ON the gateway and login to LoRa cloud page using the credential provided to know the status of the gateway and configure.

For more information contact to support@slscorp.com.